Eigenvectors and Diagonalization

Full Lecture Notes

Cleaned-up LaTeX notes from lectures dated 2019/11/05, 2019/11/07, and 2019/11/08

Contents

1 Eigenvectors and Diagonalization I **2**

1.1 The basic idea . . . 2

1.2 A motivating example: reflection in $\mathbb{R}^2$ . . . 2

1.3 Eigenvectors and eigenvalues . . . 3

1.4 Matrix version of diagonalization . . . 3

1.5 Example: rotation matrices and dependence on the field . . . 4

1.6 Characteristic polynomial of a matrix . . . 6

2 Eigenvectors and Diagonalization II **6**

2.1 Determinant and trace of a linear operator . . . 6

2.2 Eigenvalues and the determinant condition . . . 7

2.3 Linear independence of eigenvectors with distinct eigenvalues . . . 8

3 Diagonalization III **9**

3.1 Eigenspaces . . . 9

3.2 Rank-nullity and eigenspace dimension . . . 9

3.3 Example: a diagonalizable 2×2 matrix . . . 10

3.4 Example: diagonalizable with repeated eigenvalue . . . 10

3.5 Sums and direct sums of subspaces . . . 11

3.6 Diagonalization criterion using eigenspaces . . . 12

4 Summary **14**

1 Eigenvectors and Diagonalization I

1.1 The basic idea

Let V be a finite-dimensional vector space over a field $\mathbb{F}$, and let

$$T : V \to V$$

be a linear operator.

The main idea of diagonalization is this: even if T looks complicated in one basis, it may look very simple in another basis. In particular, suppose there is a basis

$$\beta = \{u_1, \dots, u_n\}$$

of V such that

$$[T]_\beta = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

Then

$$T(u_i) = d_i u_i, \qquad 1 \le i \le n.$$

Thus, with respect to the basis β, the operator T acts on each basis vector simply by scalar multiplication.

1.2 A motivating example: reflection in $\mathbb{R}^2$

Let $m \in \mathbb{R}$ and define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{1+m^2} \begin{pmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

This is the reflection across the line spanned by $(1, m)^T$.

Let

$$u_1 = \begin{pmatrix} 1 \\ m \end{pmatrix}, \qquad u_2 = \begin{pmatrix} m \\ -1 \end{pmatrix}.$$

Then $\beta = \{u_1, u_2\}$ is a basis of $\mathbb{R}^2$, and direct computation gives

$$T(u_1) = u_1, \qquad T(u_2) = -u_2.$$

Therefore

$$[T]_\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

So the same operator which may look complicated in the standard basis becomes extremely simple in the basis β.

Geometrically, the direction $\mathrm{Span}(u_1)$ is fixed by the reflection, and the perpendicular direction $\mathrm{Span}(u_2)$ is reversed.

1.3 Eigenvectors and eigenvalues

Definition 1.1 (Eigenvector and eigenvalue)**.** Let V be a finite-dimensional vector space over $\mathbb{F}$, and let $T : V \to V$ be a linear operator.

A nonzero vector $u \in V$ is called an *eigenvector* of T if there exists $\lambda \in \mathbb{F}$ such that

$$T(u) = \lambda u.$$

The scalar λ is called the *eigenvalue* corresponding to u.

Equivalently,

$$T(u) = \lambda u \quad \iff \quad (T - \lambda \,\mathrm{id}_V)(u) = 0,$$

where $u \neq 0$.

Remark 1.2. If u is an eigenvector of T with eigenvalue λ, then the one-dimensional subspace

$$\mathbb{F}u = \{cu : c \in \mathbb{F}\}$$

is T-invariant, and the restriction

$$T|_{\mathbb{F}u} : \mathbb{F}u \to \mathbb{F}u$$

acts as scalar multiplication by λ:

$$T(cu) = cT(u) = c\lambda u = \lambda(cu).$$

Definition 1.3 (Diagonalizable linear operator)**.** A linear operator $T : V \to V$ is called *diagonalizable* if there exists a basis β of V such that $[T]_\beta$ is a diagonal matrix.

Equivalently,

$$T \text{ is diagonalizable} \quad \iff \quad \text{there exists a basis of } V \text{ consisting of eigenvectors of } T.$$

Indeed, if is a basis of eigenvectors and

$$T(u_i) = \lambda_i u_i,$$

then

$$[T]_\beta = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

1.4 Matrix version of diagonalization

Let

$$A \in \mathrm{Mat}_{n\times n}(\mathbb{F})$$

and consider the linear operator

$$L_A : \mathbb{F}^n \to \mathbb{F}^n, \qquad x \mapsto Ax.$$

If L_A is diagonalizable, then there exists a basis

$$\beta = \{v_1, \ldots, v_n\}$$

of $\mathbb{F}^n$ consisting of eigenvectors of A. Suppose

$$Av_i = \lambda_i v_i.$$

Let

$$P = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}.$$

Then P is invertible and

$$AP = P\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Therefore

$$P^{-1}AP = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Definition 1.4 (Diagonalizable matrix)**.** A matrix $A \in \mathrm{Mat}_{n\times n}(\mathbb{F})$ is called *diagonalizable over* $\mathbb{F}$ if there exist an invertible matrix $P \in \mathrm{Mat}_{n\times n}(\mathbb{F})$ and a diagonal matrix D such that

$$D = P^{-1}AP.$$

Equivalently, A is similar over $\mathbb{F}$ to a diagonal matrix.

The columns of P are eigenvectors of A, and the diagonal entries of D are the corresponding eigenvalues.

1.5 Example: rotation matrices and dependence on the field

Let

$$R_\theta = L_A, \qquad A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad 0 < \theta < 2\pi, \quad \theta \neq \pi.$$

This is rotation by angle θ in the plane.

Case 1: over $\mathbb{R}$

We look for a nonzero vector $u \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$ such that

$$Au = \lambda u.$$

This is equivalent to

$$(A - \lambda I)u = 0, \qquad u \neq 0.$$

Thus $A - \lambda I$ must not be invertible, so

$$\det(A - \lambda I) = 0.$$

Compute

$$\det(A - \lambda I) = \begin{vmatrix} \cos\theta - \lambda & -\sin\theta \\ \sin\theta & \cos\theta - \lambda \end{vmatrix} = (\cos\theta - \lambda)^2 + \sin^2\theta.$$

Hence

$$\lambda^2 - 2(\cos\theta)\lambda + 1 = 0.$$

The discriminant is

$$\Delta = 4\cos^2\theta - 4 = 4(\cos^2\theta - 1) = -4\sin^2\theta < 0,$$

because $\theta \neq 0, \pi, 2\pi$. Therefore there is no real eigenvalue. Hence R_θ is not diagonalizable over $\mathbb{R}$.

Case 2: over $\mathbb{C}$

Over $\mathbb{C}$, the equation

$$\lambda^2 - 2(\cos\theta)\lambda + 1 = 0$$

has two distinct roots:

$$\lambda_1 = \cos\theta + i\sin\theta = e^{i\theta}, \qquad \lambda_2 = \cos\theta - i\sin\theta = e^{-i\theta}.$$

For $\lambda_1 = e^{i\theta}$,

$$A - \lambda_1 I = \begin{pmatrix} -i\sin\theta & -\sin\theta \\ \sin\theta & -i\sin\theta \end{pmatrix}.$$

Solving

$$(A - \lambda_1 I)\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0$$

gives

$$-iu_1 - u_2 = 0, \qquad u_2 = -iu_1.$$

So an eigenvector for λ_1 is

$$\begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Similarly, for $\lambda_2 = e^{-i\theta}$, an eigenvector is

$$\begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Thus

$$\beta = \left\{ \begin{pmatrix} 1 \\ -i \end{pmatrix}, \begin{pmatrix} 1 \\ i \end{pmatrix} \right\}$$

is a basis of $\mathbb{C}^2$, and

$$[R_\theta]_\beta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}.$$

Therefore the rotation matrix is diagonalizable over $\mathbb{C}$, but not over $\mathbb{R}$.

1.6 Characteristic polynomial of a matrix

Definition 1.5 (Characteristic polynomial)**.** Let $A \in \mathrm{Mat}_{n\times n}(\mathbb{F})$. The characteristic polynomial of A is

$$\varphi_A(t) = \det(A - tI).$$

A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A if and only if

$$\varphi_A(\lambda) = 0.$$

Exercise 1.6 (Coefficients of the characteristic polynomial)**.** Show that

$$\varphi_A(t) = (-1)^n \left(t^n + a_1 t^{n-1} + \cdots + a_{n-1}t + a_n \right),$$

where

$$a_1 = -\operatorname{tr}(A), \qquad (-1)^n a_n = \det(A).$$

Solution sketch. The constant term of $\varphi_A(t) = \det(A - tI)$ is obtained by setting $t = 0$, so it is $\det(A)$. Thus

$$(-1)^n a_n = \det(A).$$

The highest degree term comes from multiplying the diagonal terms $-t$, giving $(-t)^n = (-1)^n t^n$. The coefficient of t^{n-1} comes from choosing exactly one diagonal entry a_{ii} and choosing $-t$ from the remaining diagonal positions. Hence that coefficient is

$$(-1)^{n-1} \sum_{i=1}^{n} a_{ii} = (-1)^{n-1} \operatorname{tr}(A).$$

Comparing with

$$(-1)^n a_1 t^{n-1},$$

we get

$$a_1 = -\operatorname{tr}(A).$$

□

2 Eigenvectors and Diagonalization II

2.1 Determinant and trace of a linear operator

Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V over $\mathbb{F}$.

Definition 2.1 (Determinant and trace of a linear operator)**.** Choose any basis β of V. Define

$$\det(T) := \det([T]_\beta), \qquad \operatorname{tr}(T) := \operatorname{tr}([T]_\beta).$$

We must check that these definitions do not depend on the choice of basis.

Lemma 2.2 (Similarity preserves determinant and trace)**.** *Let $A, B \in \mathrm{Mat}_{n\times n}(\mathbb{F})$. Suppose there exists an invertible matrix P such that*

$$A = P^{-1}BP.$$

Then

$$\det(A) = \det(B), \qquad \operatorname{tr}(A) = \operatorname{tr}(B).$$

Proof. For the determinant,

$$\det(A) = \det(P^{-1}BP) = \det(P^{-1}) \det(B) \det(P) = \det(B).$$

For the trace, use the identity

$$\operatorname{tr}(XY) = \operatorname{tr}(YX)$$

for square matrices X, Y. Then

$$\operatorname{tr}(A) = \operatorname{tr}(P^{-1}BP) = \operatorname{tr}(BPP^{-1}) = \operatorname{tr}(B).$$

□

If β and γ are two bases of V, then $[T]_\beta$ and $[T]_\gamma$ are similar matrices. Therefore determinant and trace are well-defined invariants of the operator T.

2.2 Eigenvalues and the determinant condition

Recall that $u \in V$ is an eigenvector of T with eigenvalue λ if and only if

$$u \neq 0, \qquad (T - \lambda \operatorname{id}_V)(u) = 0.$$

Thus

$$\lambda \text{ is an eigenvalue} \quad \Longleftrightarrow \quad \operatorname{Ker}(T - \lambda \operatorname{id}_V) \neq \{0\}.$$

Proposition 2.3. *Let $T : V \to V$ be a linear operator on a finite-dimensional vector space. Then*

$$\lambda \in \mathbb{F} \text{ is an eigenvalue of } T \quad \Longleftrightarrow \quad \det(T - \lambda \operatorname{id}_V) = 0.$$

Proof. Choose a basis β of V. Then

$$[T - \lambda \operatorname{id}_V]_\beta = [T]_\beta - \lambda I.$$

Now

$$\operatorname{Ker}(T - \lambda \operatorname{id}_V) \neq \{0\}$$

if and only if the matrix $[T]_\beta - \lambda I$ has a nontrivial kernel. Since V is finite-dimensional, this happens if and only if the matrix is not invertible, which happens if and only if its determinant is zero. Hence

$$\lambda \text{ is an eigenvalue} \quad \Longleftrightarrow \quad \det([T]_\beta - \lambda I) = 0.$$

But this determinant is exactly $\det(T - \lambda \operatorname{id}_V)$. □

Definition 2.4 (Characteristic polynomial of a linear operator)**.** The characteristic polynomial of T is

$$\varphi_T(t) = \det(T - t \operatorname{id}_V).$$

Therefore,

$$\lambda \text{ is an eigenvalue of } T \quad \Longleftrightarrow \quad \varphi_T(\lambda) = 0.$$

2.3 Linear independence of eigenvectors with distinct eigenvalues

Lemma 2.5. *Let $u_1, \ldots, u_k$ be eigenvectors of T with eigenvalues*

$$\lambda_1, \ldots, \lambda_k.$$

Assume the eigenvalues are mutually distinct:

$$\lambda_i \neq \lambda_j \quad \text{whenever} \quad i \neq j.$$

Then

$$\{u_1, \ldots, u_k\}$$

is linearly independent.

Proof. Suppose not. Let h be the smallest positive integer such that

$$\{u_1, \ldots, u_h\}$$

is linearly dependent. Then $h \geq 2$, and $\{u_1, \ldots, u_{h-1}\}$ is linearly independent.

There exist scalars $a_1, \ldots, a_h$, not all zero, such that

$$a_1 u_1 + a_2 u_2 + \cdots + a_h u_h = 0. \tag{1}$$

By minimality of h, we must have $a_h \neq 0$.

Apply T to (1). Since $T(u_i) = \lambda_i u_i$, we get

$$a_1 \lambda_1 u_1 + a_2 \lambda_2 u_2 + \cdots + a_h \lambda_h u_h = 0. \tag{2}$$

Multiply (1) by λ_h:

$$a_1 \lambda_h u_1 + a_2 \lambda_h u_2 + \cdots + a_h \lambda_h u_h = 0. \tag{3}$$

Subtract (3) from (2):

$$a_1(\lambda_1 - \lambda_h)u_1 + \cdots + a_{h-1}(\lambda_{h-1} - \lambda_h)u_{h-1} = 0.$$

Since $u_1, \ldots, u_{h-1}$ are linearly independent, each coefficient is zero:

$$a_i(\lambda_i - \lambda_h) = 0, \qquad 1 \leq i \leq h-1.$$

Because the eigenvalues are distinct, $\lambda_i - \lambda_h \neq 0$, so

$$a_i = 0, \qquad 1 \leq i \leq h-1.$$

Then (1) becomes

$$a_h u_h = 0.$$

Since $u_h \neq 0$, we get $a_h = 0$, contradiction.

Therefore the eigenvectors are linearly independent. □

Corollary 2.6. *Let* $\dim V = n$. *If* $T : V \to V$ *has* n *distinct eigenvalues in* $\mathbb{F}$, *then* T *is diagonalizable over* $\mathbb{F}$.

Proof. Choose one eigenvector for each eigenvalue. By the lemma, these n eigenvectors are linearly independent. Since $\dim V = n$, they form a basis of V. Hence V has a basis consisting of eigenvectors, so T is diagonalizable. □

Remark 2.7. The converse is false. A linear operator may be diagonalizable even if it has fewer than n distinct eigenvalues. The identity map on an n-dimensional vector space has only one eigenvalue, 1, but it is diagonalizable.

3 Diagonalization III

3.1 Eigenspaces

Definition 3.1 (Eigenspace)**.** Let λ be an eigenvalue of $T : V \to V$. Define

$$E_\lambda = \{u \in V : T(u) = \lambda u\}.$$

This is called the *eigenspace* corresponding to λ.

Since

$$T(u) = \lambda u \quad \iff \quad (T - \lambda \operatorname{id}_V)(u) = 0,$$

we have

$$E_\lambda = \operatorname{Ker}(T - \lambda \operatorname{id}_V).$$

Therefore E_λ is a subspace of V.

Exercise 3.2. Show directly that $E_\lambda \leq V$.

Solution. Let $u, v \in E_\lambda$ and $c \in \mathbb{F}$. Then

$$T(u + v) = T(u) + T(v) = \lambda u + \lambda v = \lambda(u + v),$$

so $u + v \in E_\lambda$. Also

$$T(cu) = cT(u) = c\lambda u = \lambda(cu),$$

so $cu \in E_\lambda$. Thus E_λ is a subspace. □

3.2 Rank-nullity and eigenspace dimension

For a linear map $T : V \to W$, the first isomorphism theorem gives

$$V/\operatorname{Ker}(T) \cong \operatorname{Im}(T).$$

Taking dimensions gives

$$\dim V - \dim \operatorname{Ker}(T) = \dim \operatorname{Im}(T).$$

Therefore

$$\dim V = \operatorname{nullity}(T) + \operatorname{rank}(T).$$

This is the rank-nullity theorem.

Applying this to $T - \lambda \operatorname{id}_V$, we get

$$\dim E_\lambda = \dim \operatorname{Ker}(T - \lambda \operatorname{id}_V) = \dim V - \operatorname{rank}(T - \lambda \operatorname{id}_V).$$

The number

$$\dim E_\lambda$$

is called the *geometric multiplicity* of the eigenvalue λ.

3.3 Example: a diagonalizable 2×2 matrix

Let

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\varphi_A(t) = \det(A - tI) = \begin{vmatrix} 1-t & 1 \\ 1 & 1-t \end{vmatrix} = (1-t)^2 - 1 = t^2 - 2t = t(t-2).$$

So the eigenvalues are

$$\lambda = 0, \qquad \lambda = 2.$$

For $\lambda = 0$,

$$E_0 = \mathrm{Ker}(A) = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

For $\lambda = 2$,

$$E_2 = \mathrm{Ker}(A - 2I) = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

Thus A has two linearly independent eigenvectors in $\mathbb{R}^2$, so it is diagonalizable.

3.4 Example: diagonalizable with repeated eigenvalue

Let

$$A = \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix}.$$

Then

$$\varphi_A(t) = \det(A - tI) = \begin{vmatrix} 4-t & 0 & 1 \\ 2 & 3-t & 2 \\ 1 & 0 & 4-t \end{vmatrix}.$$

Expanding along the second column,

$$\varphi_A(t) = (3-t)\begin{vmatrix} 4-t & 1 \\ 1 & 4-t \end{vmatrix} = (3-t)((4-t)^2 - 1).$$

Thus

$$\varphi_A(t) = (3-t)(t^2 - 8t + 15) = (3-t)(t-3)(t-5) = -(t-3)^2(t-5).$$

The eigenvalues are

$$3, \qquad 5.$$

There are only two distinct eigenvalues, but A may still be diagonalizable.

For $\lambda = 5$,

$$A - 5I = \begin{pmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \\ 1 & 0 & -1 \end{pmatrix}.$$

The rank is 2, so

$$\dim E_5 = 3 - 2 = 1.$$

Solving gives

$$E_5 = \text{Span}\left\{\begin{pmatrix}1\\2\\1\end{pmatrix}\right\}.$$

For $\lambda = 3$,

$$A - 3I = \begin{pmatrix}1 & 0 & 1\\2 & 0 & 2\\1 & 0 & 1\end{pmatrix}.$$

The rank is 1, so

$$\dim E_3 = 3 - 1 = 2.$$

Solving gives

$$E_3 = \text{Span}\left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}.$$

Hence

$$\dim E_3 + \dim E_5 = 2 + 1 = 3 = \dim \mathbb{R}^3.$$

Therefore A is diagonalizable. One eigenbasis is

$$\beta = \left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}1\\2\\1\end{pmatrix}\right\},$$

and with respect to this basis,

$$[L_A]_\beta = \begin{pmatrix}3 & 0 & 0\\0 & 3 & 0\\0 & 0 & 5\end{pmatrix}.$$

3.5 Sums and direct sums of subspaces

Definition 3.3 (Sum of subspaces)**.** Let $U, W \leq V$. Define

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

This is called the sum of U and W.

Exercise 3.4. Show that

$$U + W \leq V$$

and

$$U \cap W \leq U + W.$$

Definition 3.5 (Direct sum)**.** Let $U, W \leq V$. We say

$$V = U \oplus W$$

if

$$V = U + W$$

and

$$U \cap W = \{0\}.$$

Equivalently,

$$V = U \oplus W$$

means that every vector $v \in V$ can be written uniquely as

$$v = u + w, \qquad u \in U, \quad w \in W.$$

More generally, for subspaces $W_1, \ldots, W_m \leq V$, we write

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_m$$

if every vector $v \in V$ can be written uniquely as

$$v = w_1 + w_2 + \cdots + w_m, \qquad w_i \in W_i.$$

Equivalently,

$$V = W_1 + \cdots + W_m$$

and

$$W_j \cap (W_1 + \cdots + W_{j-1} + W_{j+1} + \cdots + W_m) = \{0\}$$

for each j.

Exercise 3.6. Let $u_1, \ldots, u_n$ be a basis for V. Let

$$U = \mathrm{Span}\{u_1, \ldots, u_k\}, \qquad W = \mathrm{Span}\{u_{k+1}, \ldots, u_n\}.$$

Show that

$$V = U \oplus W.$$

3.6 Diagonalization criterion using eigenspaces

Let $T : V \to V$ be a linear operator on an n-dimensional vector space. Suppose the distinct eigenvalues of T are

$$\lambda_1, \ldots, \lambda_k, \qquad k \leq n.$$

For each i, let

$$E_{\lambda_i} = \mathrm{Ker}(T - \lambda_i \,\mathrm{id}_V).$$

Theorem 3.7 (Diagonalization theorem)**.** *The linear operator T is diagonalizable if and only if*

$$V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}.$$

Proof. First suppose T is diagonalizable. Then there exists a basis β of V consisting of eigenvectors. Group this basis according to eigenvalue:

$$\beta = \{u_{11}, \ldots, u_{1m_1}, u_{21}, \ldots, u_{2m_2}, \ldots, u_{k1}, \ldots, u_{km_k}\},$$

where each u_{ij} has eigenvalue λ_i.

Let

$$W_i = \text{Span}\{u_{i1}, \ldots, u_{im_i}\}.$$

Since β is a basis,

$$V = W_1 \oplus \cdots \oplus W_k.$$

Clearly

$$W_i \subseteq E_{\lambda_i}.$$

We claim that actually $W_i = E_{\lambda_i}$.

Let $v \in E_{\lambda_i}$. Since $V = W_1 \oplus \cdots \oplus W_k$, write uniquely

$$v = w_1 + \cdots + w_k, \qquad w_j \in W_j.$$

Then

$$T(v) = \lambda_i v.$$

But also, since T acts as multiplication by λ_j on W_j,

$$T(v) = \lambda_1 w_1 + \cdots + \lambda_k w_k.$$

Thus

$$0 = T(v) - \lambda_i v = \sum_{j=1}^{k} (\lambda_j - \lambda_i) w_j.$$

By uniqueness in the direct sum, each component must be zero. Hence for $j \neq i$,

$$(\lambda_j - \lambda_i) w_j = 0.$$

Since $\lambda_j \neq \lambda_i$, we get $w_j = 0$ for all $j \neq i$. Therefore $v = w_i \in W_i$. Thus

$$E_{\lambda_i} = W_i.$$

Hence

$$V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}.$$

Conversely, suppose

$$V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}.$$

For each i, choose a basis β_i of E_{λ_i}. Then

$$\beta = \beta_1 \cup \cdots \cup \beta_k$$

is a basis of V because the sum is direct. Every vector in β_i is an eigenvector with eigenvalue λ_i. Therefore β is a basis of V consisting of eigenvectors. Hence T is diagonalizable. □

Corollary 3.8 (Dimension criterion)**.** *Let $T : V \to V$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_k$ in $\mathbb{F}$. Then T is diagonalizable over $\mathbb{F}$ if and only if*

$$\dim E_{\lambda_1} + \cdots + \dim E_{\lambda_k} = \dim V.$$

Proof. Eigenspaces corresponding to distinct eigenvalues have zero intersection in the direct-sum sense, by the linear-independence lemma. Therefore

$$E_{\lambda_1} + \cdots + E_{\lambda_k}$$

is automatically a direct sum. Thus it equals V if and only if its dimension equals $\dim V$, namely if and only if

$$\sum_{i=1}^{k} \dim E_{\lambda_i} = \dim V.$$

□

Remark 3.9. If

$$V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_k}$$

and $m_i = \dim E_{\lambda_i}$, then with respect to a basis obtained by joining bases of the eigenspaces,

$$[T]_\beta = \begin{pmatrix} \lambda_1 I_{m_1} & & & 0 \\ & \lambda_2 I_{m_2} & & \\ & & \ddots & \\ 0 & & & \lambda_k I_{m_k} \end{pmatrix}.$$

This is a diagonal matrix whose diagonal entries are the eigenvalues repeated according to the dimensions of their eigenspaces.

4 Summary

- An eigenvector is a nonzero vector u satisfying $T(u) = \lambda u$.
- A linear operator is diagonalizable if and only if it has a basis consisting of eigenvectors.
- For a matrix A, diagonalization means finding P invertible and D diagonal such that

$$D = P^{-1}AP.$$

- The eigenvalues of A are the roots of

$$\varphi_A(t) = \det(A - tI).$$

- Eigenvectors corresponding to distinct eigenvalues are linearly independent.
- If $\dim V = n$ and T has n distinct eigenvalues, then T is diagonalizable.
- The eigenspace corresponding to λ is

$$E_\lambda = \operatorname{Ker}(T - \lambda \operatorname{id}_V).$$

- The operator T is diagonalizable if and only if

$$V = \bigoplus_{\lambda} E_\lambda,$$

  where the sum runs over the distinct eigenvalues of T.

- Equivalently,

$$T \text{ is diagonalizable} \quad \Longleftrightarrow \quad \sum_{\lambda} \dim E_\lambda = \dim V,$$

  provided all eigenvalues needed for diagonalization lie in the field $\mathbb{F}$.

- Diagonalizability depends on the field. A real rotation matrix with angle not 0 or π is not diagonalizable over $\mathbb{R}$, but it is diagonalizable over $\mathbb{C}$.